

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Fourth Semester of B. Tech. Examination (CE/IT)

Nov 2013

CE-214 MICROPROCESSOR & INTERFACES

Date: 16.11.2013, Saturday Time: 01:30 p.m. To 04:30 p.m. Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION-I

Q-1 Answer the following questions.

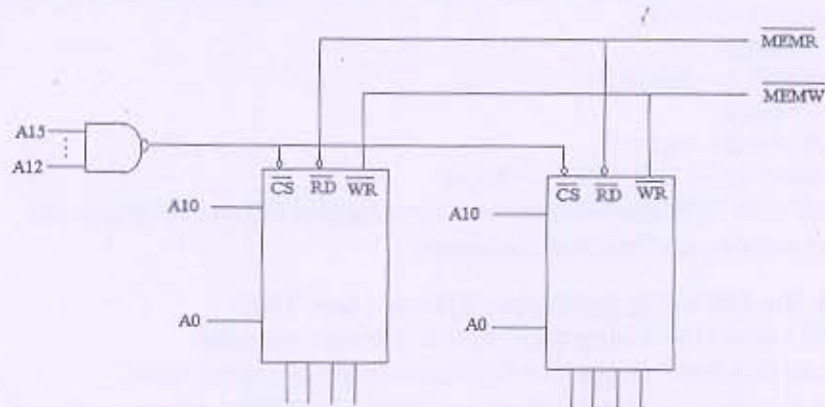
- [A] Why are the program counter and stack pointer 16-bit registers? [11]
- [B] Write the differences between PUSH-POP and CALL-RET. [2]
- [C] Compare memory mapped I/O and peripheral mapped I/O. [3]
- [D] What is use of following pins of 8085: TRAP, HLDA, INTR. [3]

Q-2 Answer the following questions.

- [A] Write an assembly program to sort the given numbers in ascending order. [12]
- [B] Write down various addressing modes of 8085 with example. [4]
- [C] What is flag? Which are the flags available in 8085 microprocessor? [4]

OR

- Q-2 [A] Write a program to count continuously in hex from 00H to FFH and FFH to 00H continuously with delay of 1.5 second between each count, and display the count at one of the output port. The T-states for various instructions are DCR-4 DCX-6 ORA-4 MVI-7 JNZ-10/7 OUT-10 MOV-4 JMP-10 [4]
- [B] Explain following instructions: XCHG, RIM [4]
- [C] Specify memory map of the following diagram and explain significance of the don't care address line on memory addresses. [4]



Q-3 Answer the following questions.(Attempt any Two)

- [A] How does 8085 perform internal data operations? How many registers it has to perform internal data operations? Write functionality of each register. [12]
- [B] What are Tri-State devices and why are they essential in a bus-oriented system?
- [C] Draw and explain Functional Block Diagram of 8085 microprocessor.
- [D] Draw and explain timing diagram of POP instruction.

SECTION-II

Q-4 Answer the following questions. [11]

- [A] If the device connected on the system bus is slow, explain the support provided by 8085 to synchronise its speed with the device speed. [2]
- [B] Explain Instruction: STA, CMP [2]
- [C] Specify the number of times the following loops are executed. [1]
XRA A
LOOP: MVI B, 55H
DCR B
JNZ LOOP
- [D] Specify the register contents and flag status(S, Z, CY) after the instruction ORA A is executed. [2]
MVI A, A9H
MVI B, 57H
ADD B
ORA A
- [E] Compare: JMP and CALL instructions in terms of function, addressing mode, type and affected flags. [2]
- [F] List various programmable peripheral devices which are compatible with intel microprocessors family. [2]

Q-5 Answer the following questions. [12]

- [A] List out the operating modes of 8253 and explain any one of them. [4]
- [B] Write an assembly language program to multiply two 8-bit numbers using successive addition method. [4]
- [C] What are the characteristics of 80386 architecture? [4]

OR

- Q-5 [A] What is Control Word of 8255? Determine the control word for the following configuration of 8255:- [4]
Port A – Output
Mode of port A – Mode 1
Port B – Output
Mode of port B – Mode 0
Port C lower (pins PC0 – PC2) – Output
- [B] Write down the differences between memory-mapped I/O and peripheral I/O. [4]
- [C] Compare software and hardware interrupts. [4]

Q-6 Answer the following questions.(Attempt any Two) [12]

- [A] Draw and explain block diagram of 8259A interrupt controller.
- [B] Explain the functional diagram of 8155 and explain its control word.
- [C] Draw and explain interfacing between 8255A and 8085 microprocessor.
- [D] Compare: 8086, 80286, 80386.